

YungChun Tu

Automation Services Leader

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Highly accomplished Software Engineer with 9+ years of experience in backend development, specializing in the .NET ecosystem, TypeScript/JavaScript, and building scalable API designs. I possess a proven track record in successful system migrations, performance optimization, microservices architecture, and comprehensive test automation. A strong cross-functional communicator (Mandarin, Japanese, English) adept at delivering pragmatic, high-impact solutions within Agile environments.

SKILLS & CERTIFICATIONS

Languages & Frameworks: C#, .NET 8/10, ASP.NET Core 3.1, .NET Framework, TypeScript/JavaScript(React, Next.js), Python, Swift, C/C++, PHP(CodeIgniter)

Architecture & Patterns: Microservices, RESTful API design, Event-driven design, Clean architecture, Domain-driven design

Data & Storage: MSSQL, MySQL, Redis

Testing & QA: MSTest, automated test suites, integration/regression testing, Locust (stress testing)

DevOps & Observability: Azure, Azure DevOps, Docker, Kubernetes, CI/CD, Prometheus, Grafana, GitHub Actions

Integration & Tools: Power Platform, Azure OpenAI, Google Maps API, Kafka, JIRA

Cloud: Azure

Languages: Mandarin (*Native*), English (*Business*), Japanese (*Business*)

WORK EXPERIENCE

AI Engineer

February 2025 - March 2026

Hitachi Astemo(TEKsystems), Tokyo, Japan

Indirect Contract

- Built internal productivity tools with Power Platform and Azure (Power Apps, Power Automate, Azure OpenAI) to automate document translation and meeting workflows.
- Developed Azure Function Apps (.NET 8) to provide reliable Translator API endpoints for Power Automate integrations.
- Migrated and integrated legacy services into a new Azure tenant, optimizing cloud architecture and supporting high-traffic workloads.
- Modernized the internal email system (SMTP MTA) by developing .NET 10 Azure Functions integrated with Azure Service Bus and Storage Account, effectively eliminating high frontend sending latency by over 60% (from 3s to ~1s).

Key technologies: Power Platform, Azure OpenAI, Azure AI Foundry, Azure Function App, .NET 8

Full-Stack Engineer

August 2024 - December 2024

BRIGHT STAR CO., LTD., Tokyo, Japan

Contractor

- Analyzed complex valve systems and translated fluid-flow calculation formulas into production-ready data models.
- Designed scalable, high-performance table schemas and indexing strategies to support intensive numeric calculations and queries.
- Implemented robust flow-calculation algorithms optimized for accuracy and throughput.
- Built real-time data processing and visualization pipelines to deliver immediate computation results for operational decision-making.

Key technologies: System architecture design, Data modeling, real-time processing, database schema & indexing

Application Engineer

May 2023 - June 2024

Rakuten(Michael Page), Tokyo, Japan

Indirect Contract

- Architected and implemented high-performance RESTful microservices on .NET 8, improving system scalability and service reliability.
- Modernized legacy .NET Framework systems through targeted refactors and modularization, increasing maintainability and performance.
- Led critical bug resolution and conducted comprehensive performance optimization, improving system reliability and user experience.
- Partnered closely with product managers throughout the software development lifecycle, from requirements gathering to feature deployment, ensuring alignment with business objectives.
- Developed automated unit and integration tests with MSTest; created Python + Locust stress tests to validate API stability under heavy load.

Key technologies: .NET 8, Microservices, REST API, MSTest, Python, Locust, CI/CD, JIRA, Kafka, Docker, Agile, Kubernetes, Prometheus, Grafana

QA Engineer

Aug 2022 - May 2023

Apple(TEKsystems), Kanagawa, Japan

Indirect Contract

- Owned QA for Siri's Chinese language support (Simplified, Traditional, and Taiwanese), ensuring correctness across dialects and environments.
- Engineered and maintained comprehensive automated test suites for Siri features using Swift 5, focusing on regression and integration testing to ensure robust functionality.
- Analyzed test results, reported defects with actionable context, and suggested product improvements to raise recognition and response quality.
- Employed Apple's testing frameworks to create expressive, maintainable test cases and improve test execution efficiency.

Key technologies: Swift 5, Apple testing frameworks, test automation, voice assistant QA, Agile

Backend Engineer

February 2020 - June 2022

Yungching Realty Co., Ltd., Taipei, Taiwan

Permanent

- Led company-wide migration from .NET Framework to ASP.NET Core 3.1 as technical lead—planned timeline, enforced quality standards, and guided cross-functional teams.
- Architected and built a modern real-estate Web API platform (ASP.NET Core) featuring 3D virtual tours, district search, and location-based services.
- Implemented Redis caching and batch-processing optimizations that improved property search performance by **50%**.
- Integrated Google Maps APIs (Geocoding API, Maps Static API, Street View Static API) to enrich property listings and map interactions.
- Optimized legacy code and managed a large-scale MSSQL database containing extensive housing data during the migration process.
- Developed comprehensive integration and unit tests using MSTest, ensuring robust API functionality and reliability throughout the system.

Key technologies: ASP.NET Core 3.1, Redis, MSSQL, Google Maps API, MSTest

Full-Stack Engineer

March 2019 - December 2019

Screea Corp., Taipei, Taiwan

Permanent

- Architected and maintained high-performance ASP.NET Core Web API 2.1 and MySQL databases for a major e-commerce platform, ensuring seamless operation under heavy traffic loads.
- Engineered a sophisticated backend system for rewards point conversion, integrating WeChat Pay to enhance payment options and user engagement.
- Partnered with UI/UX designers to deliver a responsive front-end that improved user experience and conversion metrics.
- Implemented efficient batch processing for transaction histories and automated bulk email dispatch to client finance departments, streamlining operations and improving communication.
- Contributed to strategic initiatives resulting in a **~10%** increase in registered users, demonstrating the ability to align technical solutions with business objectives.
- Developed robust web applications using PHP CodeIgniter 4, showcasing versatility across multiple programming languages and frameworks.

Key technologies: ASP.NET Core 2.1, MySQL, WeChat Pay, PHP (CodeIgniter 4)

Embedded Software Engineer

August 2016 - March 2019

Advantech Co., Ltd., Taipei, Taiwan

Permanent

- Designed and delivered C# WinForms applications for embedded systems, covering full SDLC from requirements to testing.
- Implemented robust C/C++ device drivers to improve hardware–software integration and reliability.
- Developed and maintained sophisticated WiX Toolset installers for Windows 7/10 images, streamlining software deployment processes.
- Redesigned and enhanced C# Windows Forms apps to seamlessly integrate custom features into Windows 10 environments.
- Optimized Windows images, achieving a remarkable **50%** reduction in system installation and startup time.
- Designed and implemented Node-RED flows to visualize critical hardware information on Grafana dashboards, enabling real-time monitoring of machine status on embedded computers.
- Provided expert-level helpdesk support, resolving complex customer inquiries and maintaining high satisfaction rates.

Key technologies: C#, C/C++, WiX Toolset, Node-RED, Grafana

TECHNICAL PROJECTS

Baby Beacon (Full-Stack Infant Care Platform) | [\[GitHub\]](#)

- **Overview:** Architected and developed a full-stack Progressive Web App (PWA) using Next.js 16 (App Router) and TypeScript for real-time infant activity tracking (sleep, feeding, health metrics).
- **Modern UI/UX Implementation:** Built a responsive, component-driven frontend using Tailwind CSS and shadcn/ui, ensuring accessibility and cross-platform compatibility.
- **Robust Backend & API Optimization:** Leveraged Next.js Server Actions and Prisma ORM to streamline client-server data communication, ensuring high-performance database queries and data integrity.
- **End-to-End Type Safety:** Integrated TypeScript with Zod for rigorous data validation, guaranteeing type safety from UI components down to database persistence.
- **Secure Authentication:** Implemented Better-Auth to manage secure user sessions and role-based data access for infant profiles.

Personal Portfolio & Technical Blog Platform | [\[GitHub\]](#)

- **Overview:** Architected and developed a high-performance personal platform using Next.js 16 (App Router) and TypeScript to showcase technical expertise and manage technical documentation via MDX.
- **Modern UI/UX Implementation:** Crafted a visually rich, responsive frontend using Tailwind CSS, Framer Motion, and shadcn/ui, ensuring a seamless cross-platform experience (Web/Mobile) and high accessibility.
- **Content Architecture & Performance:** Engineered an MDX-based content pipeline with custom components, leveraging Next.js features for optimized Static Site Generation (SSG) and near-instant page transitions.
- **Data Management & Type Safety:** Integrated Drizzle ORM for robust database interactions, achieving end-to-end type safety from schema definitions to UI rendering.
- **Deployment & CI/CD:** Orchestrated automated deployment pipelines via GitHub Actions, focusing on delivery speed, production reliability, and Core Web Vitals optimization.

EDUCATION

Tamkang University

2011 - 2015

Bachelor's Degree in Computer Science and Information Engineering

CERTIFICATIONS

- TOEIC® 745, December 2023
- JLPT N2, December 2023

SOFT SKILLS

- Strong analytical problem-solving, cross-functional collaboration, Agile practitioner, focus on observability and production reliability.